Programski prevodioci

02 Osnovne faze

25-26

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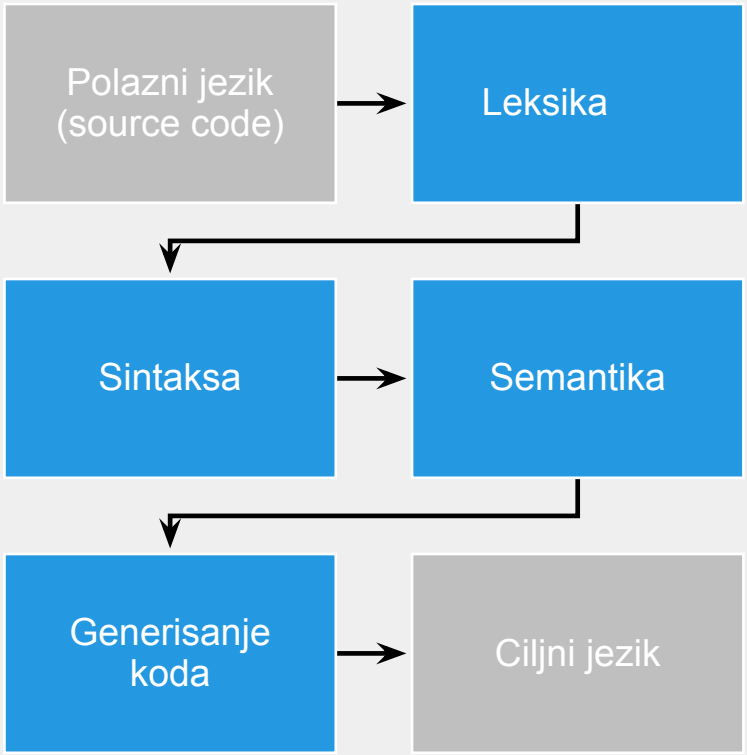
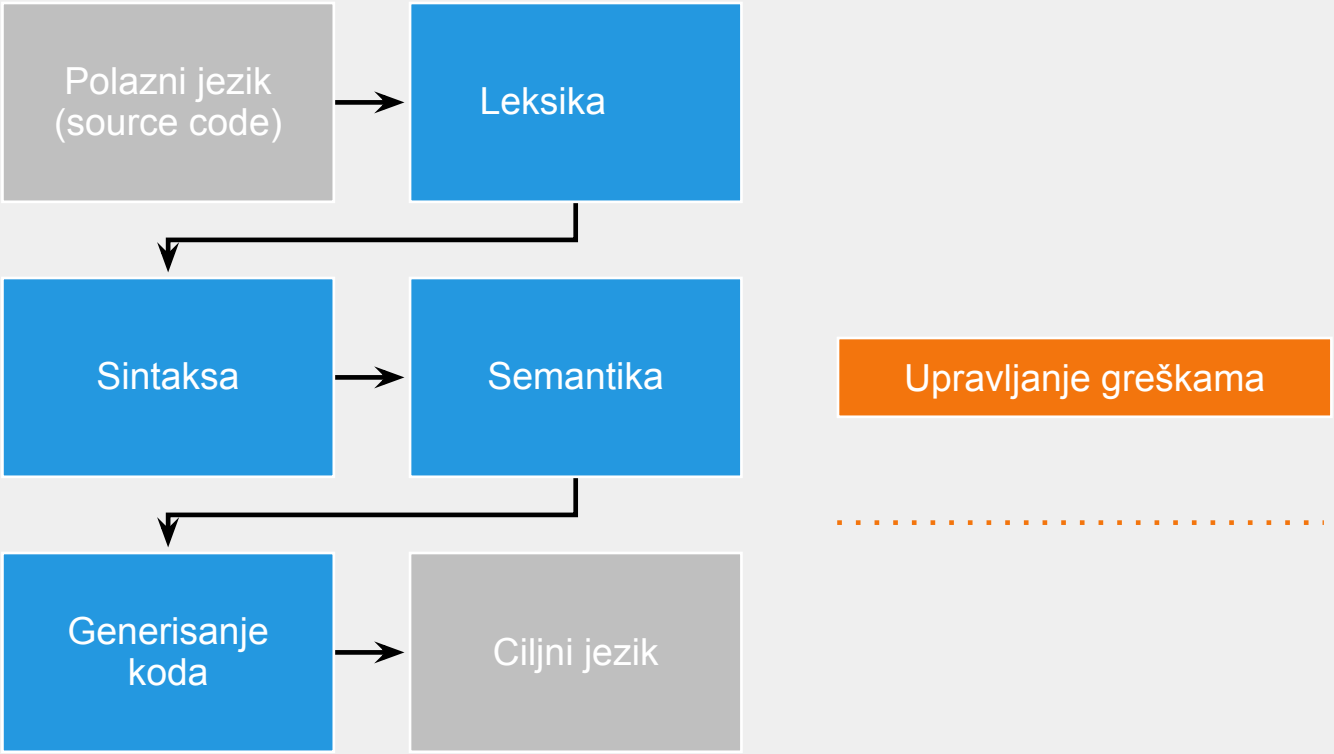

Polazni jezik
(source code)
Leksika
Sintaksa
Semantika
Generisanje
koda
Ciljni jezik


Polazni jezik
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Leksika
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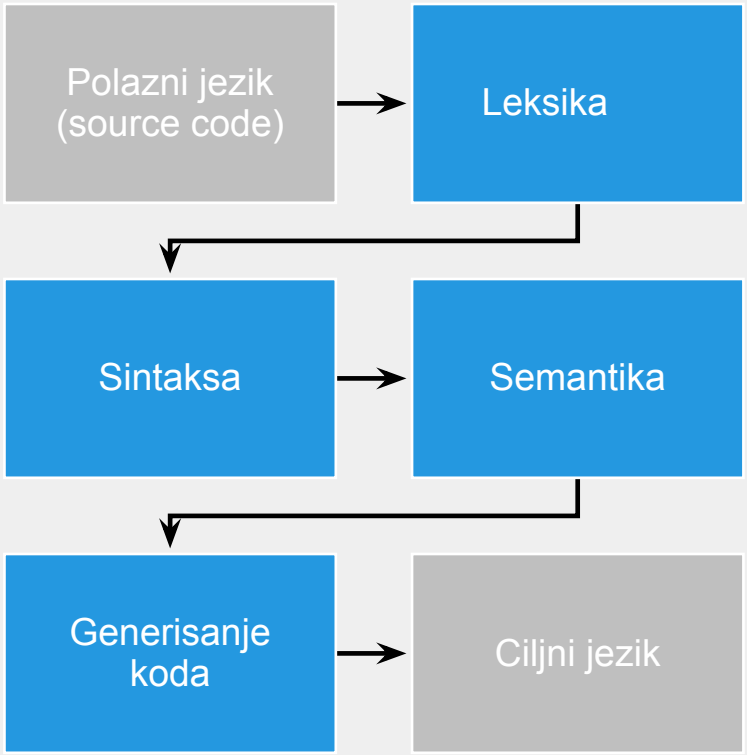
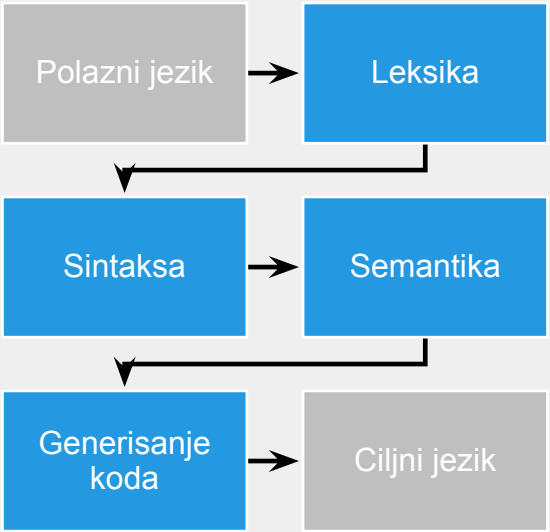
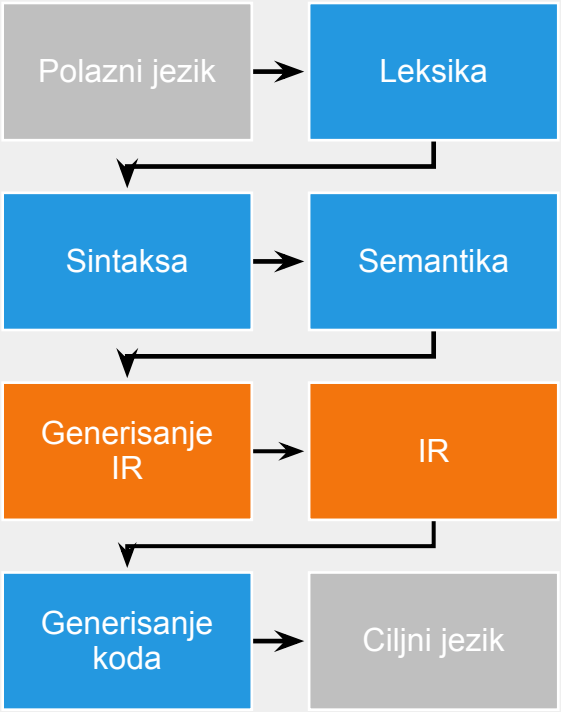
Upravljanje greškama

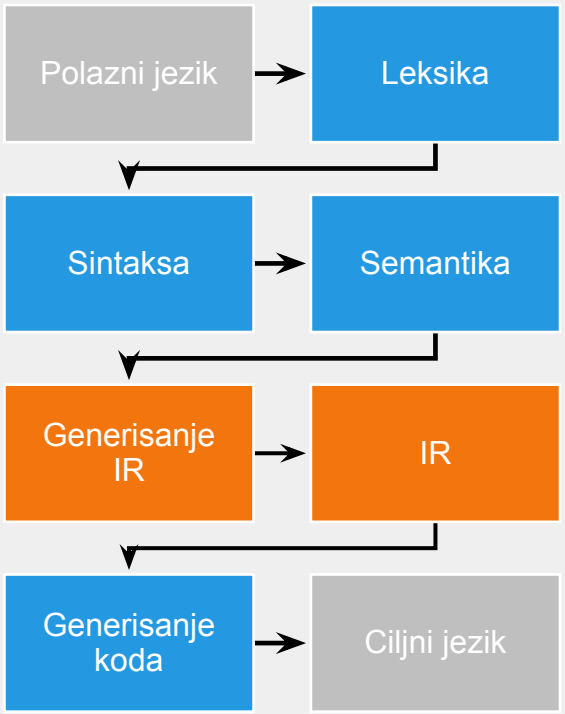


- Upravljanje greškama
- Transformacije koda
- Analiza koda
- Optimizacije
- Upravljanje resursima
- ...





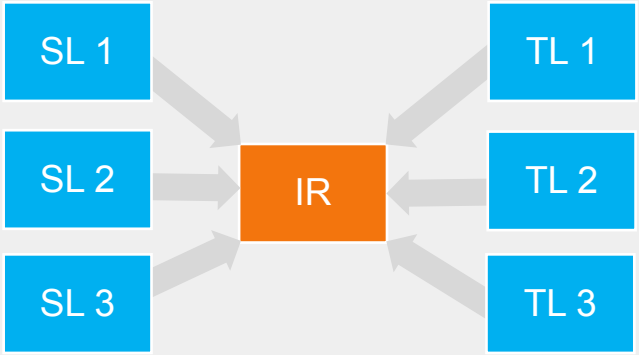
IR – Intermediate representation (međureprezentacija)



Front end

Middle end

Back end



Razvoj vašeg prevodioca

Leksička analiza - Skener

```
if (x == 5)
{
   y = 3; //komentar
}
```

if	(	x	==	5	)	{	y	=	3	;	}

lekseme

```
<KEYW, IF, 1> <LPAREN,, 1> <IDENT, x, 1>
<ROP, EQ, 1> <NUM, 5, 1> <RPAREN,, 1>
<LBRAC,, 2> <IDENT, y, 3> <ASSIGN,, 3> <NUM, 3, 3>
<SC,, 3> <RBRAC,, 4>
```

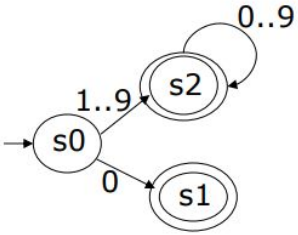
tokeni

regularni izrazi
automati
flex
C
build
test primeri

```
identifier      ->  letter ( letter | digit )*

ar_operator     ->  "+" | "-"

rel_operator    ->  "==" | "<"
```

```
SRC = scanner
.PHONY: clean

$(SRC): lex.yy.c
    gcc -o $@ $+

lex.yy.c: $(SRC).l
    flex $<

clean:
    rm -f lex.yy.c
    rm -f $(SRC).tab.c
    rm -f $(SRC).tab.h
    rm -f $(SRC)
```

```
%option noyywrap yylineno
%{
  union {
    int i;
    char *s;
  } yylval;

  enum tokens { _TYPE = 1, _IF, _ELSE, _RETURN,
                _LPAREN, _RPAREN, _LBRACKET, _RBRAC
                _SEMICOLON, _ASSIGN, _AROP, _RELOP,
                _INT_NUMBER, _UINT_NUMBER };
  char *token_strings[] = { "NONE", "_TYPE", "_IF",
```




```
int abs(int i) {
  int res;
  if(i > 0)
    res = 0 - i;
  else
    res = i;
  return res;
}

int main() {
  return abs(-5);
}
```

Sintaksna analiza - Parser

** parse tree, AST*

```
x = 5;
y 3;
```

<IDENT, x> <ASSIGN, >, <NUM, 5> - OK
<IDENT, y> , <NUM, 3> - Greška

Primer. Pretpostavljamo da je u posmatranom jeziku prva linija ispravna, a da druga nije.

gramatike
flex
bison
C
build
test primeri

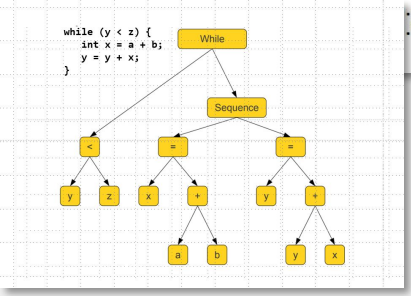
```
num_exp ::= exp
          | num_exp _ar_oper

exp     ::= literal
          | _identifier
```

```
variable_list
  : /* empty */
  | variable_list variable
  ;

variable
  : _TYPE _ID _SEMICOLON
```

```
int i = lookup_symbol($2,VAR|PAR);
if(i == NO_INDEX)
  insert_symbol($2, VAR, $1, var_num++, NO
else
  err("duplicated local var");
```

```
SRC = syntax
.PHONY: clean

$(SRC): lex.yy.c $(SRC).tab.c
    gcc -o $@ $+

lex.yy.c: $(SRC).l $(SRC).tab.c
    flex $<

$(SRC).tab.c: $(SRC).y
    bison -d $<

clean:
    rm -f lex.yy.c
```

```
int abs(int i) {
  int res;
  if(i > 0)
    res = 0 - i;
  else
    res = i;
  return res;
}

int main() {
  return abs(-5);
}
```



Semantička analiza

** AST*

```
int x = 5;
string str = “abc”;
x = str;
```

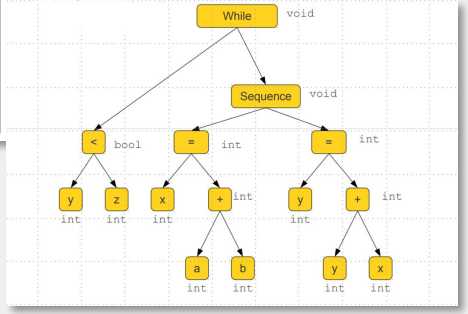
pravila semantike
flex
bison
C
build
test primeri

tabela simbola

- svi globalni identifikatori moraju biti međusobno različiti
- lokalni identifikatori raznih funkcija mogu biti identični

```
if(get_type(fun_idx) == VOID)
   err("Function cannot return value");
else if(get_type(fun_idx) != get_type($2))
   err("incompatible types in return");
return_count++;
```

STRING SIMBOLA	VRSTA SIMBOLA			
main	FUN	p		
p	PAR	parametra	parametra	-
x	VAR	tip promenljive	redni broj promenljive	-
100	LIT	tip literala	-	-
%0	REG	tip sadrzaja registra	-	-



Generisanje koda

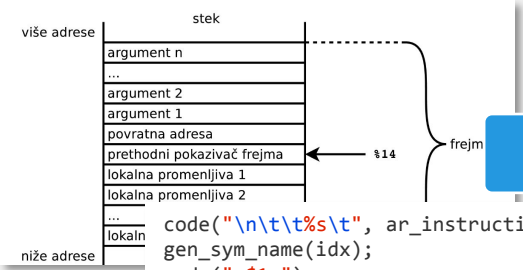
** generisanje međukoda*

```c
int f(int p) {
  int a;
  return p + a;
}
```

```
f:
   PUSH %14
   MOV %15,%14
   SUBS %15,$4,%15
@f_body:
   ADDS 8(%14),-4(%14),%0
  ...
```

asembler (hipotetski)
flex
bison
C
build
test primeri

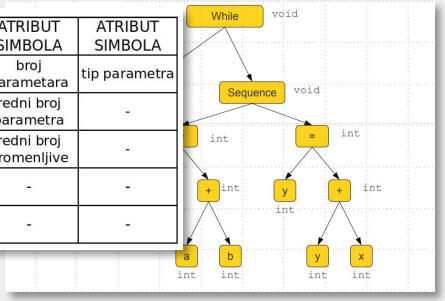
emulator



source → hip asm

```
code("\n\t\t%s\t", ar_instructions[ADD + (get_type(idx) - 1) * AROP_NUMBER]);
gen_sym_name(idx);
code(",$1,");
gen_sym_name(idx);
```

STRING SIMBOLA	VRSTA SIMBOLA	TIP SIMBOLA	ATRIBUT SIMBOLA	ATRIBUT SIMBOLA
main	FUN	povratni tip funkcije	broj parametara	tip parametra
p	PAR	tip parametra	redni broj parametra	-
x	VAR	tip promenljive	redni broj promenljive	-
100	LIT	tip literala	-	-
%0	REG	tip sadrzaja registra	-	-



hip asm → ? → target

PREDAVANJA

- podrška za vežbe (faze)
- runtime
- upravljanje memorijom
- kompajliranje, interpretiranje, JIT
- virtualne mašine, međukod
- optimizacije
- tipovi
- ...

Računarske nauke (Computer Science; CS) > Programski jezici (Programming Languages; PL)